

Differential Equations and Slope Fields

ANSWER KEY



Verifying solutions to differential equations

- 1 Verify that $y = Ce^{3x}$ is a solution to the differential equation $\frac{dy}{dx} = 3y$ for any constant C .
 $\frac{dy}{dx} = 3Ce^{3x} = 3(Ce^{3x}) = 3y$. This confirms $y = Ce^{3x}$ satisfies the equation for any C .
- 2 Verify that $y = x^2 + C$ is a general solution to $\frac{dy}{dx} = 2x$, and find the particular solution satisfying $y(1) = 5$.
 $\frac{dy}{dx} = 2x$ ✓. Using $y(1) = 5$: $1 + C = 5 \Rightarrow C = 4$. Particular solution: $y = x^2 + 4$.

Solving separable differential equations

- 3 Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 3$.
 $y dy = x dx \Rightarrow \frac{y^2}{2} = \frac{x^2}{2} + C \Rightarrow y^2 = x^2 + 2C$. Using $y(0) = 3$: $9 = 0 + 2C \Rightarrow 2C = 9$. Solution: $y^2 = x^2 + 9$, i.e. $y = \sqrt{x^2 + 9}$ (positive root, since $y(0) = 3 > 0$).
- 4 Solve $\frac{dy}{dx} = 2xy$ given $y(0) = 1$.
 $\frac{1}{y} dy = 2x dx \Rightarrow \ln|y| = x^2 + C$. Using $y(0) = 1$: $\ln 1 = 0 + C \Rightarrow C = 0$. So $\ln y = x^2 \Rightarrow y = e^{x^2}$.
- 5 Solve $\frac{dy}{dx} = \frac{\cos x}{y^2}$ given $y(0) = 2$.
 $y^2 dy = \cos x dx \Rightarrow \frac{y^3}{3} = \sin x + C$. Using $y(0) = 2$: $\frac{8}{3} = 0 + C \Rightarrow C = \frac{8}{3}$. So $y^3 = 3\sin x + 8$, i.e. $y = \sqrt[3]{3\sin x + 8}$.

Slope fields

- 6 For the differential equation $\frac{dy}{dx} = x - y$, find the slope at the points $(0, 0)$, $(1, 0)$, and $(1, 1)$, as would be drawn on a slope field.
At $(0, 0)$: slope = $0 - 0 = 0$. At $(1, 0)$: slope = $1 - 0 = 1$. At $(1, 1)$: slope = $1 - 1 = 0$.
- 7 Explain how a slope field for $\frac{dy}{dx} = ky$ (exponential growth/decay) would look different for $k > 0$ versus $k < 0$.
For $k > 0$: slopes are positive above the x -axis and negative below, steepening as $|y|$ increases — curves grow away from $y = 0$ (unstable equilibrium, exponential growth). For $k < 0$: slopes point toward $y = 0$ from both sides, flattening as $|y| \rightarrow 0$ — curves decay toward $y = 0$ (stable equilibrium, exponential decay).

Exponential growth and decay models

- 8 A bacteria culture grows according to $\frac{dP}{dt} = 0.2P$, with $P(0) = 500$. Find $P(t)$ and the population after 10 hours.
Solution form: $P(t) = P_0 e^{kt} = 500e^{0.2t}$. $P(10) = 500e^2 \approx 500(7.389) \approx 3694$ bacteria.
- 9 A radioactive substance decays according to $\frac{dA}{dt} = -0.05A$. If $A(0) = 100$ grams, find how long it takes to decay to 50 grams (the half-life).
 $A(t) = 100e^{-0.05t}$. Set $50 = 100e^{-0.05t}$: $0.5 = e^{-0.05t}$, $\ln(0.5) = -0.05t$, $t = \frac{\ln(0.5)}{-0.05} = \frac{-0.693}{-0.05} \approx 13.86$ years.

Matching slope fields to differential equations

- 10 A slope field shows horizontal segments (slope 0) along the entire x -axis and along the entire y -axis, with slopes elsewhere resembling the product xy . Identify the differential equation this slope field represents.
 $\frac{dy}{dx} = xy$ — slope is 0 whenever $x = 0$ or $y = 0$, matching the described field.

Newton's Law of Cooling

- 11 An object cools according to $\frac{dT}{dt} = -k(T - 70)$, where 70°F is room temperature. If $T(0) = 200^\circ\text{F}$ and $k = 0.1$, find $T(t)$.
Let $y = T - 70$: $\frac{dy}{dt} = -0.1y \Rightarrow y = y_0 e^{-0.1t}$. $y_0 = 200 - 70 = 130$. So $T(t) = 70 + 130e^{-0.1t}$.

Visualizing an exponential growth solution

- 12 The graph shows the bacteria population $P(t) = 500e^{0.2t}$ from the earlier problem. Use the graph to estimate how long it takes for the population to double from its initial value of 500, then verify algebraically using $\ln 2$.
The graph crosses $P = 1000$ at approximately $t \approx 3.5$. Algebraically: $1000 = 500e^{0.2t}$ gives $2 = e^{0.2t}$, so $t = \frac{\ln 2}{0.2} \approx 3.47$, matching the graphical estimate.

A logistic-style differential equation

- 13 A population satisfies $\frac{dP}{dt} = 0.1P\left(1 - \frac{P}{200}\right)$. Find the equilibrium solutions (where $\frac{dP}{dt} = 0$), and explain which one is stable for a population starting near $P = 10$.
Setting $\frac{dP}{dt} = 0$: $P = 0$ or $P = 200$. Starting near $P = 10$ (close to 0 but positive), the population will grow toward the stable equilibrium $P = 200$ (the carrying capacity), since $P = 0$ is unstable for any positive starting population.

Additional separable equations

- 14 Solve $\frac{dy}{dx} = \frac{y}{x}$ given $y(1) = 6$.

$\frac{1}{y}dy = \frac{1}{x}dx \Rightarrow \ln|y| = \ln|x| + C \Rightarrow y = Ax$ for a constant A . Using $y(1) = 6$: $A = 6$: $y = 6x$.

- 15 Solve $\frac{dy}{dx} = \frac{1}{y}$ given $y(0) = 4$.

$y dy = dx \Rightarrow \frac{y^2}{2} = x + C$. Using $y(0) = 4$: $8 = 0 + C \Rightarrow C = 8$. So $y^2 = 2x + 16$, i.e. $y = \sqrt{2x + 16}$ (positive root, since $y(0) = 4 > 0$).

A mixing problem modeled by a differential equation

- 16 A tank contains 50 L of pure water. Brine with concentration 3 kg/L flows in at 2 L/min, and the well-mixed solution drains at the same rate. If $y(t)$ is the amount of salt (kg) at time t , the differential equation is $\frac{dy}{dt} = 6 - \frac{2y}{50} = 6 - \frac{y}{25}$. Find the equilibrium value of y , and explain what it represents physically. Setting $\frac{dy}{dt} = 0$: $6 = \frac{y}{25}$ gives $y = 150$ kg. This represents the long-run salt content: eventually the rate of salt entering (6 kg/min from the inflow) balances the rate leaving, keeping the amount constant at 150 kg.